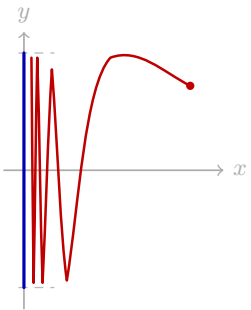

Figure 1: Topologist's sine curve: $y = \sin(1/x)$ accumulates on $\{0\} \times [-1, 1]$; connected, not path-connected.